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Thermodynamic Modeling of the Ag-X (X = B, Fe, Sm, Pu) Binary Systems

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Abstract Based on the experimental data available in the literature, the Ag-X (X = B, Fe, Sm, Pu) binary systems were assessed by means of the CALPHAD (CALculation of PHAse Diagrams) approach. Six intermetallic compounds, i.e. αAg_2Sm, βAg_2Sm, AgSm, $Ag_{51}Sm_{14}$, $Ag_{51}Pu_{14}$ and Ag_2Pu, were treated as stoichiometric phases. The solution phases including liquid, (Ag, γFe, δPu), (βB), (αFe, δFe, εPu), (αSm, γSm), (βSm), (αPu), (βPu), (γPu) and (δ'Pu) were described by the substitutional solution model with the Redlich–Kister polynomial. A set of self-consistent thermodynamic parameters for each of the binary systems was obtained. The calculated phase diagrams and thermodynamic properties were in agreement with the reported experimental data.

Keywords Ag-X (X = B, Fe, Sm, Pu) · CALPHAD · phase diagram · thermodynamic modeling

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1 Introduction

Due to the high thermal stability, excellent conductivity, and good oxidation and corrosion resistance, silver and its alloys are used extensively in the form of circuit board, conductive element, heat conductive element, metal plating, supercapacitor or nanopaste etc., in the electrical and electronic industries.[1-3] Ag with low strength cannot serve in a high-temperature environment for a long time. Fortunately, the strength and performance of Ag-based alloys at elevated temperatures can be significantly improved by adding a small amount of alloying elements.[4-6] It is well known that the addition of a small amount of B element into alloys and the surface-protection coatings may significantly improve their mechanical properties and high-temperature stability due to the interstitial solid solution strengthening.[7-9] Fe and Ag form complete immiscibility even in the liquid phase indicating a strong repulsive interaction between the two elements. The Ag-Fe alloys have been used as antimicrobial surfaces for biodegradable stents[10] or antibacterial activity[11] as well as bimetallic AgFe catalysts.[12] Sm element has the large thermal neutron capture cross-section making it valuable as luminescent and in IR-absorbing glasses.[13] Pu is one of the actinides formed by neutron capture in U with the fact that 5f electrons sit at the boundary between localization and itinerancy. Besides, Pu and its alloys are attractive materials with many anomalous physical properties for use in nuclear fuels and nuclear weapons.[14,15] The Ag-X (X = S, As, Lu, Ce, Dy, Er) systems[6,16,17] have been assessed thermodynamically, but not for the Ag-X (X = B, Fe, Sm, Pu) systems. The detailed information of phase equilibria and thermodynamic properties of the Ag-X (X = B, Fe, Sm, Pu) binary systems is essential to develop and design the related alloys.

Therefore, the purposes of the present work are to critically evaluate the experimental phase equilibria and thermodynamic data for the Ag-X (X = B, Fe, Sm, Pu) binary systems available in the literature and to obtain a set of self-consistent thermodynamic parameters for each of binary systems by means of the CALPHAD approach.

2 Evaluation of Experimental Data in the Literature

To facilitate reading, the experimental data on the phase diagram and thermodynamic properties of the Ag-X (X = B, Fe, Sm, Pu) systems are briefly summarized in Table 1. The crystal structures of solid phases in these binary systems are listed in Table 2.

2.1 The Ag-B System

The phase equilibria information for the Ag-B system are rather limited. Only one piece of experimental data for the monotectic reaction Liquid#2 → Liquid#1 + (βB) at 962 °C from Wald[18] exists. Obrowski[19] studied the structure of compound AgB_2 by direct synthesis at an unspecified temperature based on x-ray diffraction (XRD). Later, Wall and Stormont[20] investigated alloys with 66.6 at.% B annealed at 900 °C for 60 days by the XRD, optical metallography (OM) and thermal analysis (TA). The results show that the compound AgB_2 does not exist. Wald and Stormont[20] also reported the almost complete immiscibility in the liquid and the solid phase using the microscope methods. This is in agreement with the earliest report by Gieblehausen.[21] Then, Wald[18] determined the Ag-B phase diagram above 900 °C based on the results of that Ag and B were immiscible in both the liquid and the solid phases, and extrapolated the monotectic reaction Liquid# 2 → Liquid#1 + (βB) at the melting point of B. Karakaya and Thompson[22] reviewed the Ag-B system, but not provided the phase diagram. Consequently, the experimental data from Wald[18] are accepted and the so-called AgB_2 phase is metastable and not included in the present optimization. All the crystal structures for solid phases in the Ag-B system are summarized in Table 2 according to literature data.[19,23,24]

2.2 The Ag-Fe System

Petrenko[25] first studied the phase equilibria of the Ag-Fe system. It was reported that the mutual solubility of Fe and Ag is extremely low in both the liquid and the solid phases and found essentially no changes in the liquid/solid transformation temperatures when mixing the two metals together. No intermetallic compound was found in this system. Experimental observation of the solubility for the Fe-rich alloys from Gibson and Rothery[26] are consistent with the ones reported by Petrenko.[25] Chipman and Floridis[27] roughly calculated the solubility of Fe in (Ag) to be about 0.2-0.4 at.% and Ag in (δFe) to be about 0.2-0.7 at.% at 1600 °C. Tamman and Oelsen[28] investigated the solubility of Fe in (Ag) to be from 4 to 6 ppm (part per million) in weight within the temperature range from 1000 to 1600 °C. However, the experimental results determined by Tamman and Oelsen[28] do not seem to be reliable enough in view of the existence of impurities in the samples. Later, Bernardini[29] performed some experiments using higher purity of alloys and reported a higher solubility of Fe in (Ag) to be from 2 to 80 ppm in weight between 650 and 912 °C, based on a radioactive tracer diffusion measurement. Subsequently, Wriedt et al.[30] determined the solubility of Ag in (γFe) on the Fe-rich side to be from 37 to 135 at-ppm in the temperature range from 1093 to 1288 °C using a combination of spark-emission spectroscopy and isopiestic technique. Besides, Swartzendruber[31] reviewed the Ag-Fe system and calculated the phase diagram over the entire composition range utilizing the computerized optimization procedure of Sundman.[32] The monotectic reaction Liquid#1 → Liquid#2 + (δFe), the peritectic reaction Liquid#2 + (δFe) → (γFe), the eutectic reaction Liquid#2 → (γFe) + (Ag), and the eutectoid reaction (γFe) → (αFe) + (Ag) were calculated to be at 1533 °C and 168 at-ppm Fe, 1398 °C and 89 at-ppm Fe, 961 °C and 16 at-ppm Fe, as well as 912 °C and 2 at-ppm Fe, respectively. However, the lattice stabilities of pure elements Ag and Fe utilized in the work of Swartzendruber[31] are inconsistent with the ones[33] in the present work.

Thus, the Ag-Fe system should be reassessed by the CALPHAD approach. Considering the experimental data from Chipman and Floridis,[27] Bernardini,[29] Wriedt et al.[30] and Swartzendruber[31] are consistent with each other, their data were thus utilized in the present optimization. The crystal structures for solid phases in the Ag-Fe system are summarized in Table 2 according to literature data.[24,34]

2.3 The Ag-Sm System

The characteristics of the Ag-Sm system are similar to those of the Ag-Ce system.[16] Gebhardt et al.[35] initially determined the Ag-Sm phase diagram over the whole composition range employing OM and XRD. In their work, three compounds Ag_3Sm (melting point 910 °C), Ag_2Sm (peritectic formation at 770 °C), and AgSm (melting point 960 °C) have been reported. Moreover, three eutectic reactions, Liquid → (Ag) + Ag_3Sm, Liquid → Ag_2Sm + Ag_3Sm and Liquid → AgSm + (Sm), were

Table 1 Summary of phase diagram and thermodynamic data for Ag-X (X = B, Fe, Sm, Pu) systems available in the literature

System	Method(a)	Type of data	QM(b)	References
Ag-B	XRD	The crystal structure of Ag_2B	–	19
	XRD, TA, OM	Confirm that Ag_2B does not exist	+	20
	…	Liquid#1 → (Ag) + (βB) at 962 °C	+	18
		Liquid# 2 → Liquid#1 + (βB) at melting point of B	+	18
Ag-Fe	EC	Phase diagram data at 1600 °C	+	27
	RTD	Solid solubility of Fe in (Ag)	+	29
	SES, WM, IT	Solid solubility of Ag in (γFe)	+	30
	EC	Phase diagram data	■	31
Ag-Sm	OM, BRXT, TA	Melting point of Ag_3Sm and AgSm	+	35
	OM, BRXT, TA	Melting point of Ag_2Sm	–	35
	OM, BRXT, TA	Liquid → Ag_3Sm + Ag_2Sm at 750 °C	–	35
	OM, BRXT, TA	Liquid → (Ag) + $Ag_{51}Sm_{14}$ at 775 °C	+	35
	OM, BRXT, TA	Liquid → AgSm + (αSm) at 690 °C	–	35
	GXRM	Homogeneity range of Ag_3Sm and AgSm	–	36
	XRD, OM	The crystallographic data for $Ag_{51}Sm_{14}$	+	38
	XRD, OM, TA	Homogeneity range of $Ag_{51}Sm_{14}$	■	39
	XRD, OM, TA	Solid solubility of Sm in (Ag)	■	39
	XRD, OM, TA	Liquid → (Ag) + $Ag_{51}Sm_{14}$ at 781 °C	■	39
	TA, XRD, OM	Melting point of Ag_3Sm and AgSm	+	40
	TA, XRD, OM	Liquid → (Ag) + $Ag_{51}Sm_{14}$ at 760 °C	+	40
		Liquid → βAg_2Sm + AgSm at 775 °C		
		Liquid → AgSm + (αSm) at 675 °C		
	TA, XRD, OM	$Ag_{51}Sm_{14}$ + Liquid → βAg_2Sm at 790 °C	+	40
	TA, XRD, OM	αAg_2Sm → βAg_2Sm at 694 °C	+	40
	TA, XRD, OM	Liquidus at the congruent melting point of AgSm phase	■	40
	IC	Enthalpy of mixing at 1233 °C	+	43
	HTDSC	Enthalpy of formation of AgSm, Ag_2Sm, $Ag_{51}Sm_{14}$	+	42
	IAS	Enthalpy of formation of AgSm, Ag_2Sm, $Ag_{51}Sm_{14}$	■	13
	IC	Enthalpy of mixing of liquid at 1177-1233 °C	+	13
Ag-Pu	XRD	The crystallographic data for $Ag_{51}Sm_{14}$	+	47
	XRD, OM, TA	Melting point of $Ag_{51}Pu_{14}$	+	46
	XRD, OM, TA	Liquid#1 → Liquid# 2 + Ag_2Pu at 795 °C	+	46
	XRD, OM, TA	Liquid#1 → (Ag) + $Ag_{51}Pu_{14}$ at 801 °C	+	46
	XRD, OM, TA	Liquid# 2 + $Ag_{51}Pu_{14}$ → Ag_2Pu at 817 °C	+	46

(a) *XRD* x-ray diffraction, *TA* thermal analysis, *OM* optical metallography, *RTD* radioactive tracer diffusion, *SES* spark-emission spectroscopy, *IT* isopiestic technique, *EC* Extrapolation calculation, *OM* microscope methods, *BRXT* back-reflection x-ray technology, *GXRM* Guinier x-ray method, *CA* chemical analysis, *HTDSC* high temperature direct synthesis calorimetry, *IC* isoperibolic calometry, extract technology, *IAS* ideal associated solutions

(b) *QM* quoted Mode, indicating whether the data are used in the parameter: + used; – not used; ■ not used but estimated to be reliable data for checking the modeling

determined to occur at 775 °C and 11.5 at.% Sm, 750 °C and 36 at.% Sm, as well as 690 °C and 76 at.% Sm, respectively. Steeb and Godel[36] determined only two intermediate phases, i.e., Ag_3Sm and AgSm, using the Guinier XRD and a high-temperature camera to 730 °C. Steeb et al.[37] studied the structure of the Ag_3Sm phase and gave the lattice parameters based on the Guinier XRD. Then, McMasters et al.[38] reported the existence of the $Ag_{51}Sm_{14}$ phase to be the Ag-rich phase using a combination of XRD and optical metallography. McMasters et al.[38] and Gschneidner et al.[39] also confirmed the existence of $Ag_{51}Sm_{14}$ by means of XRD, TA and OM. These authors[38,39] pointed out that the Ag_3Sm phase reported by groups of investigators[35-37] was actually

Table 2 List of the symbols to denote the phases in the Ag-X (X = B, Fe, Sm, Pu) systems

Phase	Prototype	Pearson symbol	Space group	Phase description	References
(βB)	βB	*hR105*	$R\bar{3}m$	Solid solution based on beta_Rhomb B	23
(Ag)	Cu	*cF4*	$Fm\bar{3}m$	Solid solution based on fcc Ag	24
(αFe)	W	*cI2*	$Im\bar{3}m$	Solid solution based on bcc Fe	24
(γFe)	Cu	*cF4*	$Fm\bar{3}m$	Solid solution based on fcc Fe	34
(δFe)	W	*cI2*	$Im\bar{3}m$	Solid solution based on bcc Fe	34
(αSm)	αSm	*hR3*	$R\bar{3}$m	Solid solution based on Rhomb Sm	41
(βSm)	Mg	*hP2*	$P6_3/mmc$	Solid solution based on hcp Sm	41
(γSm)	W	*cI2*	$Im\bar{3}m$	Solid solution based on bcc Sm	41
αAg_2Sm	…	…	…	Low-temperature form of the Ag_2Sm	40
βAg_2Sm	Ag_3Pu	*hP(?)*	*P63(*)*	High-temperature form of the Ag_2Sm	40
AgSm	CsCl	*cP2*	$Pm\bar{3}m$	Binary phase AgSm	37
$Ag_{51}Sm_{14}$	$Ag_{51}Gd_{14}$	*hP65*	*P6/m*	Solid solution based on $Ag_{51}Sm_{14}$	37
(αPu)	αPu	*mP16*	$P2_1/m$	Solid solution based on αPu	51
(βPu)	βPu	*mC34*	*I2/m(a)*	Solid solution based on βPu	52
(γPu)	γPu	*oM8*	*Fddd*	Solid solution based on γPu	49
(δPu)	Cu	*cF4*	$Fm\bar{3}m$	Solid solution based on fcc Pu	53
(δ'Pu)	In	*tI2*	*I4/mmm*	Solid solution based on Tetragonal Pu	50
(εPu)	W	*cI2*	$Im\bar{3}m$	Solid solution based on bcc Pu	53
$Ag_{51}Pu_{14}$	$Ag_{51}Gd_{14}$	…	$P6_322$	Binary phase $Ag_{51}Pu_{14}$	48
Ag_2Pu	…	…	…	Binary phase Ag_2Pu	46

$Ag_{51}Sm_{14}$ with slight solubility range due to the similar lattice parameters. Their results[38,39] were considered in the present work. In addition, they reported the maximum solubility of Sm in (Ag) was 0.42 at.% and determined the eutectic reaction Liquid → (Ag) + $Ag_{51}Sm_{14}$ to be at 781 °C which is higher than the values from Gebhardt et al.[35] Stapf et al.[40] constructed the phase diagram over the entire composition range using a combination of XRD, TA and OM. According to their work, three stoichiometric compounds Ag_3Sm (melting point 935 °C), βAg_2Sm (peritectic formation at 790 °C), and AgSm (melting point 960 °C) have been reported. Three eutectics reactions, Liquid → (Ag) + Ag_3Sm, Liquid → Ag_2Sm + AgSm and Liquid → AgSm + (Sm), were determined to be at 760 °C and 10 at.% Sm, 775 °C and 37 at.% Sm, as well as 675 °C and 82 at.% Sm, respectively. The peritectic reaction, Liquid + Ag_3Sm → βAg_2Sm was determined to be at 790 °C and 33 at.% Sm. Besides, the Ag_2Sm phase was determined in two allotropic forms: αAg_2Sm and βAg_2Sm, and the transformation reaction of αAg_2Sm → βAg_2Sm occurs at 694 °C. Then, based on the above reported experimental data, Gschneidner and Calderwood[41] constructed the phase diagram of the Ag-Sm system over the whole composition. Sudavtsova et al.[13] calculated the liquidus of the phase diagram based on ideal associated solutions model, but not provided the thermodynamic parameters.

Four groups of the researchers[13,42-44] studied the thermodynamic properties data of the Ag-Sm system. Niessen et al.[42] predicted the standard enthalpies of formation at 25 °C of AgSm, Ag_2Sm and $Ag_{51}Sm_{14}$ to be − 42, − 40 and − 28 in kJ/mole-atoms based on semi-empirical model, respectively. Ivanov and Witusiewicz[43] reported the enthalpies of mixing for melts at 1233 °C in the range of 0-46 at.% Sm based on isoperibolic calorimetry. Meschel and Kleppa[44] measured the standard enthalpies of formation of AgSm, Ag_2Sm and $Ag_{51}Sm_{14}$ to be − 37.0 ± 2.7, − 24.1 ± 2.4 and − 16.8 ± 3.5 in kJ/mole of atoms, respectively, using the high temperature direct synthesis calorimetry at 1000 ± 2 °C. Subsequently, Sudavtsova et al.[13] determined the enthalpies of mixing for liquid based on isoperibolic calorimetry at 1177-1233 °C in the range of 46-100 at.% Sm. Sudavtsova et al.[13] also calculated the activities of the pure components and the molar fractions of associates in melts of the Ag-Sm system. Moreover, the enthalpies of formation of AgSm and $Ag_{51}Sm_{14}$ are also reported in the Open Quantum Materials Database (OQMD).[45]

According to the experimental results,[13,35,36,38-40] the $Ag_{51}Sm_{14}$, Ag_2Sm and AgSm phases were treated as stoichiometric compounds in the present work. The

experimental results[40,44,45] were utilized in the present work due to the consistency among them. All the structure information[37,40,41] was summarized in Table 2.

2.4 The Ag-Pu System

The experimental contribution on the phase diagram over the entire composition range of the Ag-Pu system is mainly from Wood et al.[46] Schonfeld[47] first reported that the mutual solubility of Pu and Ag extremely low in the liquid phase, which is in agreement with the data from Runnalls.[48] Later, Wood et al.[46] reported two stoichiometric compounds, i.e. $Ag_{51}Pu_{14}$ (melting point 977 °C) and Ag_2Pu (peritectic formation at 817 °C) based on a combination of XRD, thermal analysis and microscope methods. Furthermore, the eutectic reaction Liquid#1 → (Ag) + $Ag_{51}Pu_{14}$, the monotectic reaction Liquid#1 → Liquid#2 + Ag_2Pu, as well as the two peritectic reactions Liquid#2 + $Ag_{51}Pu_{14}$ → Ag_2Pu and Liquid#2 + Ag_2Pu → (εPu) were also determined by Wood et al.[46] to be at 801 ± 2 °C and 9.2 at.% Pu, 795 ± 5 °C and 41 at.% Pu, 817 ± 5 °C and 659 ± 5 °C, respectively. The Ag_3Pu phase was firstly found by Runnalls[48] based on XRD, then it was confirmed to be the $Ag_{51}Pu_{14}$ phase reported by Wood et al. [46] The thermodynamic properties for $Ag_{51}Pu_{14}$ and Ag_2Pu phases were not available in the open literature. The phase equilibria data reported by Wood et al.[46] were utilized in the present thermodynamic optimization. The crystal structure data in the Ag-Pu system are given in Table 2 according to literature data.[46,49-53] The crystal structure of the Ag_2Pu phase was not reported.

3 Thermodynamic Modeling

3.1 Pure Elements

The thermodynamic properties of pure elements Ag, B, Fe, Sm and Pu are taken from the SGTE database compiled by Dinsdale.[33] The Gibbs energy function ${}^{\circ}G_i^{\varphi} = G_i^{\varphi} - H_i^{\mathrm{SER}}$ for pure element i (i = Ag, B, Fe, Sm and Pu) in the phase φ is described by the following equation:

$$ {}^{\circ}G_i^{\varphi} = a + bT + cT\ln(T) + dT^2 + eT^{-1} + fT^3 + gT^7T + hT^{-9} \quad \text{(Eq 1)} $$

in which H_i^{SER} represents the molar enthalpy of the elements in their reference states at 298.15 K and 1 bar. The coefficients a, b, c, d, e, f, g and h are constant, and T is absolute temperature. The last two terms in Eq. 1 are used only for the metastable ranges, gT^7 for a liquid below the melting point and hT^{-9} for solid phases above the melting point.

3.2 Solution Phases

In the Ag-X (X = B, Fe, Sm, Pu) systems, the Gibbs energies of the solution phases, i.e. liquid, (Ag, γFe, δPu), (βB), (αFe, δFe, εPu), (αSm, γSm), (βSm), (αPu), (βPu), (γPu) and (δ'Pu), are described by the substitutional solution model with the Redlich–Kister polynomial.[54] The molar Gibbs energy of the solution phases is expressed as follows:

$$ G_m^{\varphi} - H^{\mathrm{SER}} = \sum_{i=\mathrm{Ag,X}} {}^{\circ}G_i^{\varphi} x_i + RT \sum_{i=\mathrm{Ag,X}} x_i \ln x_i + x_{\mathrm{Ag}} x_{\mathrm{X}} \sum_{j=0}^{n} {}^{j}L_{\mathrm{Ag,X}}^{\varphi} \left(x_{\mathrm{Ag}} - x_{\mathrm{X}}\right)^j \quad \text{(Eq 2)} $$

in which H^{SER} is the abbreviation of $x_{\mathrm{Ag}} H_{\mathrm{Ag}}^{\mathrm{SER}} + x_{\mathrm{X}} H_{\mathrm{X}}^{\mathrm{SER}}$, R is the gas constant, x_{Ag} and x_{X} are the mole fractions of Ag and X (X= B, Fe, Sm, Pu), respectively. $\sum_{i=\mathrm{Ag,X}} {}^{\circ}G_i^{\varphi} x_i$ represents the mechanical mixing of the pure elements to the Gibbs energy, RT $\sum_{i=\mathrm{Ag,X}} x_i \ln x_i$ denotes the contribution of the ideal entropy of mixing to the Gibbs energy, and ${}^{j}L_{\mathrm{Ag,X}}^{\varphi}$ is the jth Redlich–Kister parameter of solution phase φ, and typically ${}^{j}L_{\mathrm{Ag,X}}^{\varphi} = a_j + b_j T$. The coefficients a_j and b_j are optimized according to the experimental data in the present work.

3.3 Intermetallic Compounds

In view of experimentally observed negligible homogeneity range,[35,36,38-40,46] the six compounds (αAg_2Sm, βAg_2Sm, AgSm, $Ag_{51}Sm_{14}$, $Ag_{51}Pu_{14}$ and Ag_2Pu) are treated as stoichiometric compounds. The Gibbs energy of each compound per mole-formula, taking Ag_2Pu as an example, can be expressed as

$$ G_m^{\mathrm{Ag_2Pu}} - 2H_{\mathrm{Ag}}^{\mathrm{SER}} - H_{\mathrm{Pu}}^{\mathrm{SER}} = a^{\mathrm{Ag_2Pu}} + b^{\mathrm{Ag_2Pu}} T + 2{}^{\circ}G_{\mathrm{Ag}}^{\mathrm{fcc}} + {}^{\circ}G_{\mathrm{Pu}}^{\mathrm{Tetragonal}} \quad \text{(Eq 3)} $$

Where $a^{\mathrm{Ag_2Pu}}$ and $b^{\mathrm{Ag_2Pu}}$ are the parameters to be optimized in the present work. ${}^{\circ}G_{\mathrm{Ag}}^{\mathrm{fcc}}$ and ${}^{\circ}G_{\mathrm{Pu}}^{\mathrm{Tetragonal}}$ are the mole Gibbs energy of Ag and Pu in their reference states. Analogous expression to Eq. 3 is applied to describe the Gibbs energies of the αAg_2Sm, βAg_2Sm, $Ag_{51}Sm_{14}$, AgSm, and $Ag_{51}Pu_{14}$ phases.

Table 3 Assessed thermodynamic parameters for the Ag-X (X = B, Fe, Sm, Pu) systems

System	Phases/models	Thermodynamic parameters
Ag-B	liquid: (Ag, B)$_1$	${}^0L^{\text{liquid}}_{\text{Ag,B}} = 208,287 - 46.685 \cdot T$
	(Ag): (Ag, B)$_1$Va$_1$	${}^0L^{(\text{Ag})}_{\text{Ag,B}} = 44,484$
Ag-Fe	liquid: (Ag, Fe)$_1$	${}^0L^{\text{liquid}}_{\text{Ag,Fe}} = 81,088$
		${}^1L^{\text{liquid}}_{\text{Ag,Fe}} = -8801$
	fcc: (Ag, Fe)$_1$Va$_1$	${}^0L^{\text{fcc}}_{\text{Ag,Fe}} = 113,281$
	bcc: (Ag, Fe)$_1$ Va$_3$	${}^0L^{\text{bcc}}_{\text{Ag,Fe}} = 124,715$
	hcp: (Ag, Fe)$_1$ Va$_{0.5}$	${}^0L^{\text{hcp}}_{\text{Ag,Fe}} = 102,788$
Ag-Sm	liquid: (Ag, Sm)$_1$	${}^0L^{\text{liquid}}_{\text{Ag,Sm}} = -80,722 - 3.179 \cdot T$
		${}^1L^{\text{liquid}}_{\text{Ag,Sm}} = -33,247 + 2.470 \cdot T$
		${}^2L^{\text{liquid}}_{\text{Ag,Sm}} = 7544 + 3.981T$
	αAg$_2$Sm: (Ag)$_{2/3}$(Sm)$_{1/3}$	${}^\circ G^{\alpha\text{Ag}_2\text{Sm}}_{\text{Ag:Sm}} = -25614 + 1.471 \cdot T + 2/3 \cdot {}^\circ G^{\text{fcc}}_{\text{Ag}} + 1/3 \cdot {}^\circ G^{\text{Rhomb}}_{\text{Sm}}$
	βAg$_2$Sm: (Ag)$_{2/3}$(Sm)$_{1/3}$	${}^\circ G^{\beta\text{Ag}_2\text{Sm}}_{\text{Ag:Sm}} = -20,441 - 3.879 \cdot T + 2/3 \cdot {}^\circ G^{\text{fcc}}_{\text{Ag}} + 1/3 \cdot {}^\circ G^{\text{Rhomb}}_{\text{Sm}}$
	AgSm: (Ag)$_{1/2}$(Sm)$_{1/2}$	${}^\circ G^{\text{AgSm}}_{\text{Ag:Sm}} = -27,784 - 0.065 \cdot T + 1/2 \cdot {}^\circ G^{\text{fcc}}_{\text{Ag}} + 1/2 \cdot {}^\circ G^{\text{Rhomb}}_{\text{Sm}}$
	Ag$_{51}$Sm$_{14}$: (Ag)$_{51/65}$(Sm)$_{14/65}$	${}^\circ G^{\text{Ag}_{51}\text{Sm}_{14}}_{\text{Ag:Sm}} = -21,923 - 0.367 \cdot T + 51/65 \cdot {}^\circ G^{\text{fcc}}_{\text{Ag}} + 14/65 \cdot {}^\circ G^{\text{Rhomb}}_{\text{Sm}}$
Ag-Pu	liquid: (Ag, Pu)$_1$	${}^0L^{\text{liquid}}_{\text{Ag,Pu}} = 557 - 5.613 \cdot T$
		${}^1L^{\text{liquid}}_{\text{Ag,Pu}} = -407,720 + 3.854 \cdot T$
	Ag$_{51}$Pu$_{14:}$ (Ag$_{51/65}$Pu$_{14/65}$)	${}^\circ G^{\text{Ag}_{51}\text{Pu}_{14}}_{\text{Ag:Pu}} = -7618 - 3.642 \cdot T + 51/65 \cdot {}^\circ G^{\text{fcc}}_{\text{Ag}} + 14/65 \cdot {}^\circ G^{\text{Tetragonal}}_{\text{Pu}}$
	Ag$_2$Pu: (Ag)$_{2/3}$(Pu)$_{1/3}$	${}^\circ G^{\text{Ag}_2\text{Pu}}_{\text{Ag:Pu}} = -12163 - 0.772 \cdot T + 2/3 \cdot {}^\circ G^{\text{fcc}}_{\text{Ag}} + 1/3 \cdot {}^\circ G^{\text{Tetragonal}}_{\text{Pu}}$

Temperature (T) in Kelvin, Gibbs energy in J/mol-atom and pressure (P) in Pa of formula unit. The Gibbs energies for the pure elements are derived from the compilation of Dinsdale[33]

4 Results and Discussion

The PARROT module based on the Thermo-Calc software and the step-by-step optimization procedure described by Du et al.[55] were used to optimize the thermodynamic parameters in the present work. Experimental phase diagram and thermodynamic data from literature were employed and each experimental data point was given a certain weight during the optimization process.

Taking the Ag-Sm system as an example, the optimization started with AgSm phase. The enthalpy of formation of AgSm from the OQMD database was used as a starting value for evaluation of the parameter a^{AgSm} in Eq. 3. Then, the parameters a_0, b_0, a_1 and b_1 for the liquid phase in Eq. 2)were adjusted to reproduce the experimental enthalpy of mixing and liquidus data. The parameter b^{AgSm} in Eq. 3 was evaluated to describe the invariant reaction Liquid → AgSm + (αSm). Similar to the AgSm phase, the parameters of the αAg$_2$Sm, βAg$_2$Sm and Ag$_{51}$Sm$_{14}$ phases were introduced one by one. It was found that the parameters a_0, b_0, a_1 and b_1 for the liquid phase in Eq. 2 were not enough to reproduce the temperatures of most of invariant reactions. The differences between calculated and experimental temperatures were more than 20 °C for some invariant reactions. Thus, the parameters a_2 and b_2 for the liquid phase in Eq. 2 were included in the present optimization. Finally, all the parameters were optimized simultaneously using all the selected experimental data. The optimization procedures for the Ag-X (X = B, Fe, Pu) systems were analogue to that for the Ag-Sm system. The finally obtained parameters for the Ag-X (X = B, Fe, Sm, Pu) systems are listed in Table 3.

Figure 1 presents the calculated Ag-B phase diagram over the entire composition compared with the experimental data.[18] Table 4 presents a comparison of the calculated invariant reactions and literature data for the Ag-B system. As can be seen from Fig. 1 and Table 4, all the reliable experimental data were reproduced well by the present work. The calculated critical point of the liquid miscibility gap locates at 50 at.% Sm and 3017 °C. The calculated temperature of the eutectic reaction Liquid#1 → (Ag) + (βB) is to be 964 °C, which is consistent with the experimental one 962 °C.

Figure 2(a) shows the calculated phase diagram of the Ag-Fe system along with the results of Chipman and Floridis,[27] Bernardini,[29] Wriedt et al.[30] and

Swartzendruber.[31] Figure 2(b) and (c) are the enlargements of Ag-rich and Fe-rich sides of the Ag-Fe phase diagram along with the experimental data.[27,30] Table 5 presents a comparison of the calculated invariant reactions and literature data for the Ag-Fe system. As can be seen from Fig. 2(a) and Table 5, the calculated results are consistent with the literature data. The differences between the calculated and reported temperatures are less than 1 °C. The calculated monotectic reaction Liquid#1 → Liquid#2 + (δFe), the peritectic reaction Liquid#2 + (δFe) → (γFe), the eutectic reaction Liquid#2 → (γFe) + (Ag) and the eutectoid reaction (γFe) → (αFe) + (Ag) are calculated to be at 1533 °C and 99.9847 at.% Fe, 1389 °C and 99.9919 at.% Fe, 961 °C and 99.9984 at.% Fe, as well as 912 °C and 99.9998 at.% Fe, respectively. These calculated results are consistent with the literature data[31] (1533 °C and 99.9832 at.% Fe, 1389 °C and 99.9911 at.% Fe, 961 °C and 99.9984 at.% Fe, as well as 912 °C and 99.9998 at.% Fe, respectively). The calculated solubility of Fe in (Ag) is about 0.00101 at.% Fe at 912 °C in Fig. 2(b), which is slightly less than the value of experiment 0.00764 at.% Fe. In view of the solubility of Fe in (Ag) is very low, the calculation value is acceptable. Besides, the calculated critical point of the liquid miscibility gap in the Ag-Fe system locates at 4726 °C and 58 at.% Fe.

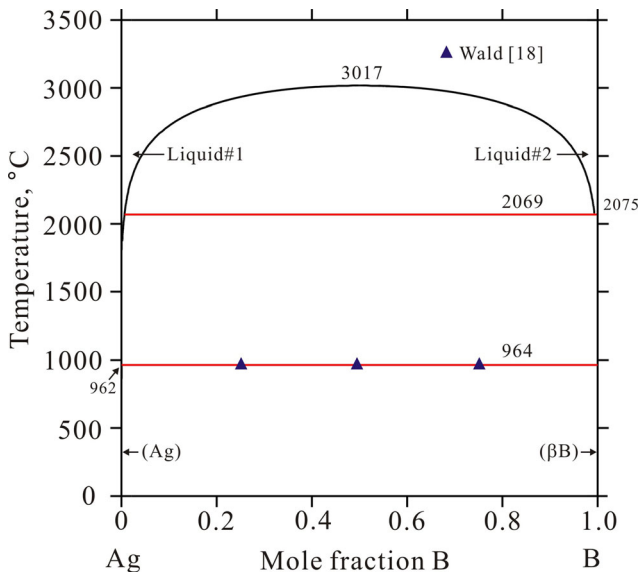

Fig. 1 Calculated Ag-B phase diagram compared with experimental data[18]

Table 4 Calculated invariant reactions in the Ag-B system compared with the experimental data

Type	Invariant reactions	Temperature, °C	Composition, at.% B			References
Monotectic	Liquid# 2 → Liquid#1 + (βB)	2069	99.34	0.66	100.00	This work
		2075	…	…	…	18
Eutectic	Liquid#1 → (Ag) + (βB)	964	0.15	0.00	100.00	This work
		962	…	0.00	100.00	18

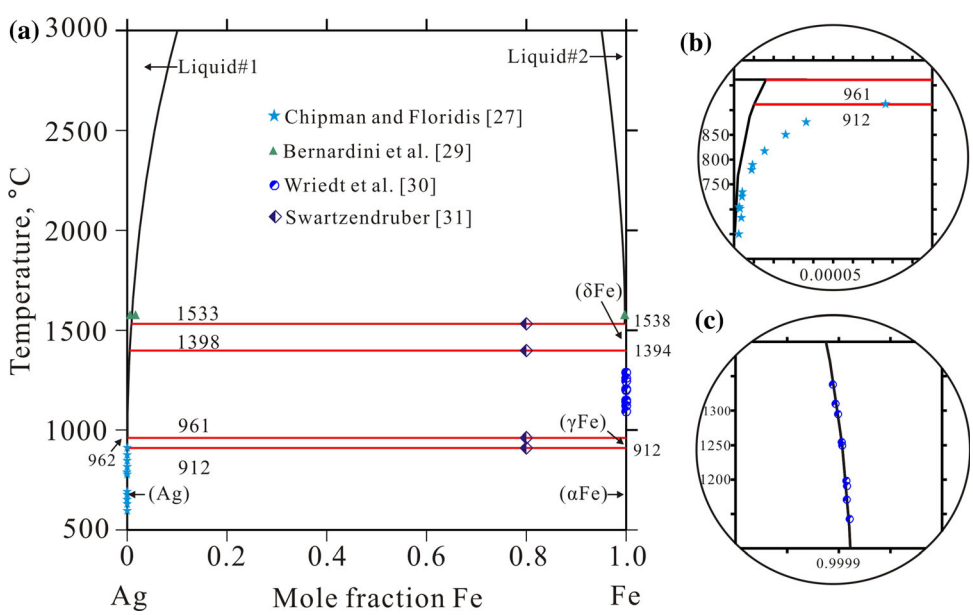

Fig. 2 Calculated Ag-Fe phase diagram: (a) the whole composition compared with experimental data,[27,29-31] (b) the enlargement of the phase diagram in the Ag-rich side from 600 to 1000 °C and (c) the enlargement of the phase diagram in the Fe-rich side from 1100 to 1400 °C

Table 5 Calculated invariant reactions in the Ag-Fe system compared with the experimental data

Type	Invariant reactions	Temperature, °C	Composition, at.% Fe			References
Monotectic	Liquid#1 → Liquid#2 + (δFe)	1533	99.9830	99.7400	0.9000	This work
		1533	99.9847	99.7412	0.8625	31
Peritectic	Liquid#2 + (δFe) → (γFe)	1398	99.9910	99.9780	0.5000	This work
		1398	99.9919	99.9785	0.5323	31
Eutectic	Liquid#2 → (γFe) + (Ag)	961	99.9984	0.05780	0.0016	This work
		961	99.9984	0.06000	0.0070	31
Eutectiod	(γFe) → (αFe) + (Ag)	912	99.9998	99.9999	0.0010	This work
		912	99.9998	99.9990	0.0040	31

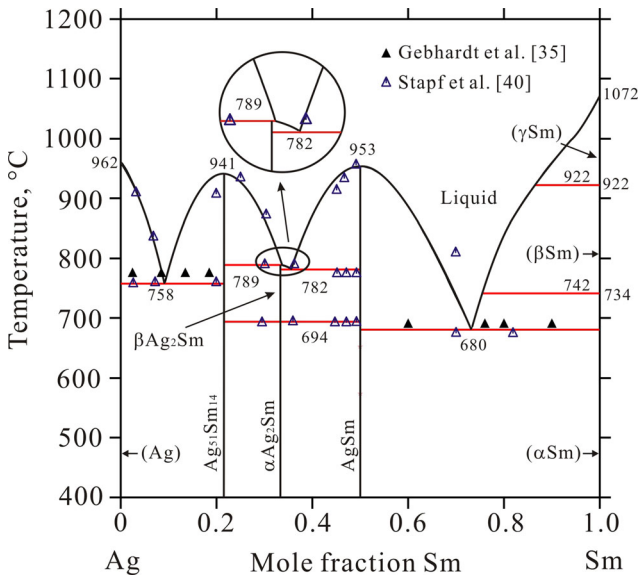

Fig. 3 Calculated Ag-Sm phase diagram compared with experimental data[35,40]

The calculated Ag-Sm phase diagram with the experimental data[35,40] is shown in Fig. 3. Table 6 presents a comparison of the calculated invariant reactions and literature data for the Ag-Sm system. It can be observed from Fig. 3 and Table 6, the calculated results are in agreement with the experimental data from Stapf et al.[40] The deviations between the calculated invariant reaction temperatures and experimental ones by Stapf et al.[40] are within 8 °C. The calculated AgSm and $Ag_{51}Sm_{14}$ phases melt congruently at 953 and 941 °C, which agree well with the experimental temperatures 960 and 935 °C, respectively. The two compounds αAg_2Sm and βAg_2Sm are formed via the allotropic and peritectic reactions at 694 °C and 782 °C, which agree well with the experimental temperatures 694 and 789 °C, respectively. According to the experimental phase diagram,[40] it is worth noting that the liquidus of the AgSm phase is asymmetrical. According to the thermodynamic principles and thermodynamic characteristics reported by Okamoto and Massalski,[56,57] the reported liquidus of the AgSm phase from Stapf et al.[37] is questionable. Hence, the composition of the eutectic point in the silver-rich side reported by Stapf et al.[40] was assigned a low weight in the present work. Similarly, the liquidus of the $Ag_{51}Sm_{14}$ phase from Stapf et al.[37] is also asymmetrical. The similar situation occurs in the work of Sudavtsova et al.[13] Thus, the calculated liquidus of the AgSm and $Ag_{51}Sm_{14}$ phases are acceptable.

Figure 4 presents the calculated enthalpies of formation at room temperature in the Ag-Sm system along with the experimental data,[44] first-principles calculations,[45] and the ones from Sudavtsova et al.[13] in dot-dashed line and Niessen et al.[42] in dashed line. The calculated enthalpies of formation in this work are in good agreement with the first-principles calculated values[45] and experimental data[44] except for the values for $Ag_{51}Sm_{14}$ phase. For the $Ag_{51}Sm_{14}$ phase, the calculated enthalpy of formation is − 21.9 kJ/mole-atoms in the present work, which is lower than the reported one − 16.8 ± 3.5 kJ/mole-atoms.[44] The experimental temperatures of invariant reactions Liquid → (Ag) + $Ag_{51}Sm_{14}$, Liquid + $Ag_{51}Sm_{14}$ → βAg_2Sm and the enthalpy of mixing in the liquid phases data cannot be reproduced if the experimental enthalpy of formation of the $Ag_{51}Sm_{14}$ phase[44] was utilized in the optimization. The same situation also occurs for AgSm phase. If we try to fit the experimental data − 37 ± 2.7 kJ/mole-atoms of the AgSm phase,[44] the calculated temperature of invariant reaction Liquid → AgSm + (αSm) is 20 °C lower than experimental data from Stapf et al.,[40] and the enthalpy of mixing of the liquid phase cannot be fitted well. Actually, it is not possible to simultaneously describe well both the enthalpy of mixing at 1233 °C from Ivanov and Witusiewicz,[43] and Sudavtsova et al.[13] as well as the experimental enthalpy of formation from Meschel and Kleppa.[41] Moreover, the experimental enthalpies of mixing reported by Ivanov and Witusiewicz,[43] and Sudavtsova et al.[13] are more

Table 6 Calculated invariant reactions in the Ag-Sm system compared with the experimental data

Type	Invariant reactions	Temperature, °C	Composition, at.% Sm			References
Eutectic	Liquid → (Ag) + $Ag_{51}Sm_{14}$	757	9.20	0.00	21.50	This work
		760	10.00	<2.0 0	…	40
		775	11.50	…	…	35
		781	…	…	…	39
Peritectic	Liquid + $Ag_{51}Sm_{14}$ → βAg_2Sm	789	21.50	33.60	33.30	This work
		790	…	36.00	…	40
Eutectic	Liquid → βAg_2Sm + AgSm	782	35.60	33.30	50.00	This work
		775	37.00	…	…	40
Eutectic	Liquid → AgSm + (αSm)	680	74.30	50.00	100.00	This work
		675	82.00	…	…	40
		690	76.00	…	…	35
Congruent	Liquid → AgSm	953	50.00			This work
		960	50.00			40
		960	…			35
Congruent	Liquid → $Ag_{51}Sm_{14}$	941	21.50			This work
		935	…			40
		910	…			35
Allotropic	αAg_2Sm → βAg_2Sm	694	33.30			This work
		694				40

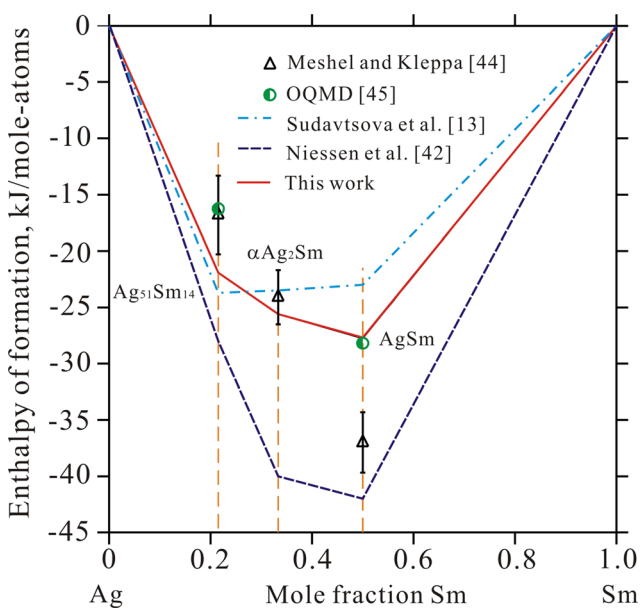

Fig. 4 Calculated enthalpies of formation at room temperature in the Ag-Sm system compared with experimental data[13,42,44,45]

consistent with each other than those of the formation enthalpy of the $Ag_{51}Sm_{14}$ and AgSm phases.[13,42,44,45] If the value of the enthalpy of mixing was utilized in this optimization, the experimental phase diagram can be reproduced well. Besides, if the value of the enthalpy of formation (-28.2 kJ/mole-atoms) for the AgSm phase calculated by the first-principles calculations[45] was accepted in the present work, the most of experimental data associated with the AgSm phase are reproduced well. Thus, the experimental phase diagram data and the enthalpy of mixing in the liquid were reproduced at the expense of the experimental enthalpy of formation of the $Ag_{51}Sm_{14}$ and AgSm phases. Table 7 is a comparison of the calculated enthalpies of formation of compounds and literature data for the Ag-Sm system.[13,42,44,45] Figure 5 shows the calculated enthalpy of mixing of the liquid phases at 1233 °C along with the experimental data from Ivanov and Witusiewicz,[43] and Sudavtsova et al. [13] As shown in this figure, the present calculated results are in good agreement with the experimental data.

The calculated Ag-Pu phase diagram in comparison with the experimental values[46] is presented in Fig. 6. The calculated melting temperature of $Ag_{51}Pu_{14}$ is 980 °C, which is consistent with reported value 977 °C.[46] The two peritectic reactions Liquid#2 + $Ag_{51}Pu_{14}$ → Ag_2Pu and Liquid#2 + Ag_2Pu → (εPu) are calculated to be at 817 and 659 °C, which are consistent with literature value 817 ± 5 and 659 ± 5 °C, respectively. The five peritectoid reactions Ag_2Pu + (εPu) → (δ'Pu), Ag_2Pu + (δ'Pu) → (δPu), Ag_2Pu + (δPu) → (γPu), Ag_2Pu + (γPu) → (βPu) and Ag_2Pu + (βPu) → (αPu) are calculated to be at 476, 463, 322, 215 and 125 °C, which are consistent with literature value 483, 463, 320, 215 and 125 °C, respectively. The eutectic reaction Liquid#1 → (Ag) + $Ag_{51}Pu_{14}$ and the monotectic reaction Liquid#1 → Liquid#2 + Ag_2Pu are calculated to be at 789 and 793 °C, which agree well with the experimental results of 801 ± 2 and 795 ± 5 °C, respectively. Furthermore, the

Table 7 Summary of experimental and calculated enthalpies of formation of compounds in Ag-Sm system

Phase	$\Delta_f H$, kJ/mole-atoms	Temperature, °C	Method	References
AgSm	− 42.0	25	Evaluation	42
	− 37.0 ± 2.7	25	HTDSC	44
	− 23.0	25	IAS	13
	− 28.2 (Pm3m)	− 273	First-principles calculations	45
	− 27.8	25	CALPHAD	This work
αAg_2Sm	− 40.0	25	Evaluation	42
	− 24.1 ± 2.4	25	HTDSC	44
	− 23.5	25	IAS	13
	− 25.6	25	CALPHAD	This work
$Ag_{51}Sm_{14}$	− 28.0	25	Evaluation	42
	− 16.8 ± 3.5	25	HTDSC	44
	− 23.7	25	IAS	13
	− 15.8 (P63/mmc)	− 273	First-principles calculations	45
	− 21.9	25	CALPHAD	This work

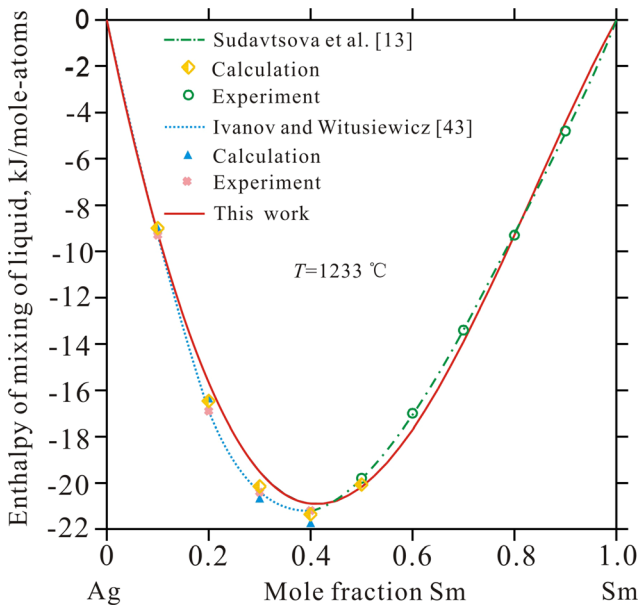

Fig. 5 Calculated enthalpies of mixing for the liquid at 1233 °C in the Ag-Sm system compared with experimental data[13,43]

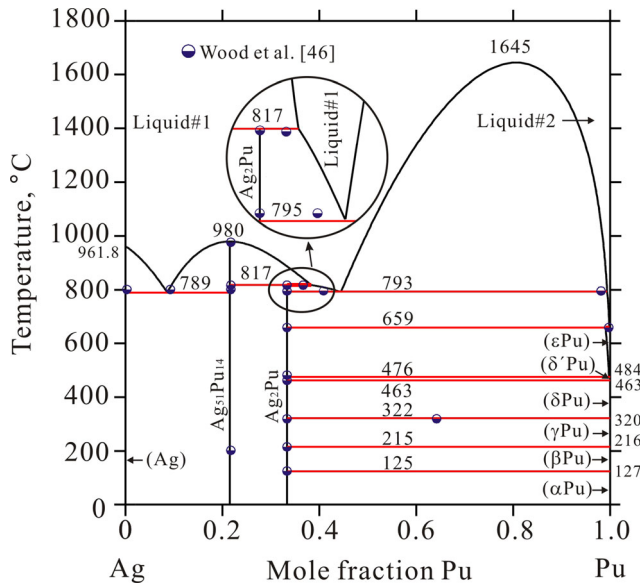

Fig. 6 Calculated Ag-Pu phase diagram compared with experimental data[46]

calculated critical point of the liquid miscibility gap locates at 1645 °C and 80.6 at.% Pu. Table 8 summarizes the calculated invariant reactions in the Ag-Pu systems compared with the experimental data.[46]

The thermodynamic parameters for the Ag-X (X = B, Fe, Sm, Pu) systems obtained in the present optimization can enrich thermodynamic database for multicomponent Ag alloys to provide a reliable basis for thermodynamic extrapolations to related ternary and higher-order systems.

5 Summary

All the experimental phase diagram and thermodynamic data available in the literature for the Ag-X (X = B, Fe, Sm, Pu) systems were critically evaluated. A set of self-consistent thermodynamic parameters for the Ag-X (X = B, Fe, Sm, Pu) systems was obtained using the CALPHAD approach. The calculated phase diagram and thermodynamic properties agree satisfactorily with the reliable experimental data.

Table 8 Calculated invariant reactions in the Ag-Pu system compared with the experimental data

Type	Invariant reactions	Temperature, °C	Composition, at.% Pu			References
Eutectic	Liquid#1 → (Ag) + $Ag_{51}Pu_{14}$	789	8.63	0.00	21.54	This study
		801 ± 2	9.20	…	21.55	46
Peritectic	Liquid# 2 + $Ag_{51}Pu_{14}$ → Ag_2Pu	817	38.36	21.54	33.33	This work
		817 ± 5	…	21.55	31.30	46
Monotectic	Liquid#1 → Liquid# 2 + Ag_2Pu	793	44.51	99.38	33.33	This work
		795 ± 5	41.00	98.00	31.30	46
Peritectic	Liquid# 2 + Ag_2Pu → (εPu)	659	99.83	33.33	99.03	This work
		659 ± 5	…	31.30	…	46
Peritectoid	Ag_2Pu + (εPu) → (δ'Pu)	476	33.33	99.76	100.00	This work
		483	31.30	…	…	46
Peritectoid	Ag_2Pu + (δ'Pu) → (δPu)	463	33.33	100.00	100.00	This work
		463				46
Peritectoid	Ag_2Pu + (δPu) → (γPu)	322	33.33	99.94	100.00	This work
		320	31.30			46
Peritectoid	Ag_2Pu + (γPu) → (βPu)	215	33.33	99.99	99.99	This work
		215	31.30			46
Peritectoid	Ag_2Pu + (βPu) → (αPu)	125	33.33	99.98	100.00	This work
		125	31.30			46
Congruent	Liquid#1 → $Ag_{51}Pu_{14}$	980	21.54			This work
		977	21.55			46

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